

1. Expansion & Special Products

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Key Algebraic Identities

$$iGrades 1/2 (a + b)iGrades 1/2 = aiGrades 1/2 + 2ab + biGrades 1/2$$

$$iGrades 1/2 (a - b)iGrades 1/2 = aiGrades 1/2 - 2ab + biGrades 1/2$$

$$iGrades 1/2 \text{ Difference of Two Squares: } aiGrades 1/2 - biGrades 1/2 = (a - b)(a + b)$$

$$iGrades 1/2 (a + b)iGrades 1/2 = aiGrades 1/2 + 3aiGrades 1/2b + 3abiGrades 1/2 + biGrades 1/2$$

Expanding Polynomials

Use the distributive law for multiplying expressions:

$$(2x + 3)(3x - 4) = 6xiGrades 1/2 - 8x + 9x - 12 = 6xiGrades 1/2 + x - 12.$$

2. Factorization Techniques

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Factorization by Grouping

For 4-term expressions: $ax + ay + bx + by = a(x + y) + b(x + y) = (a + b)(x + y)$.

Simplifying Algebraic Fractions

To simplify algebraic fractions, factorize numerator and denominator completely, then cancel common factors. Never cancel individual terms across addition or subtraction signs.